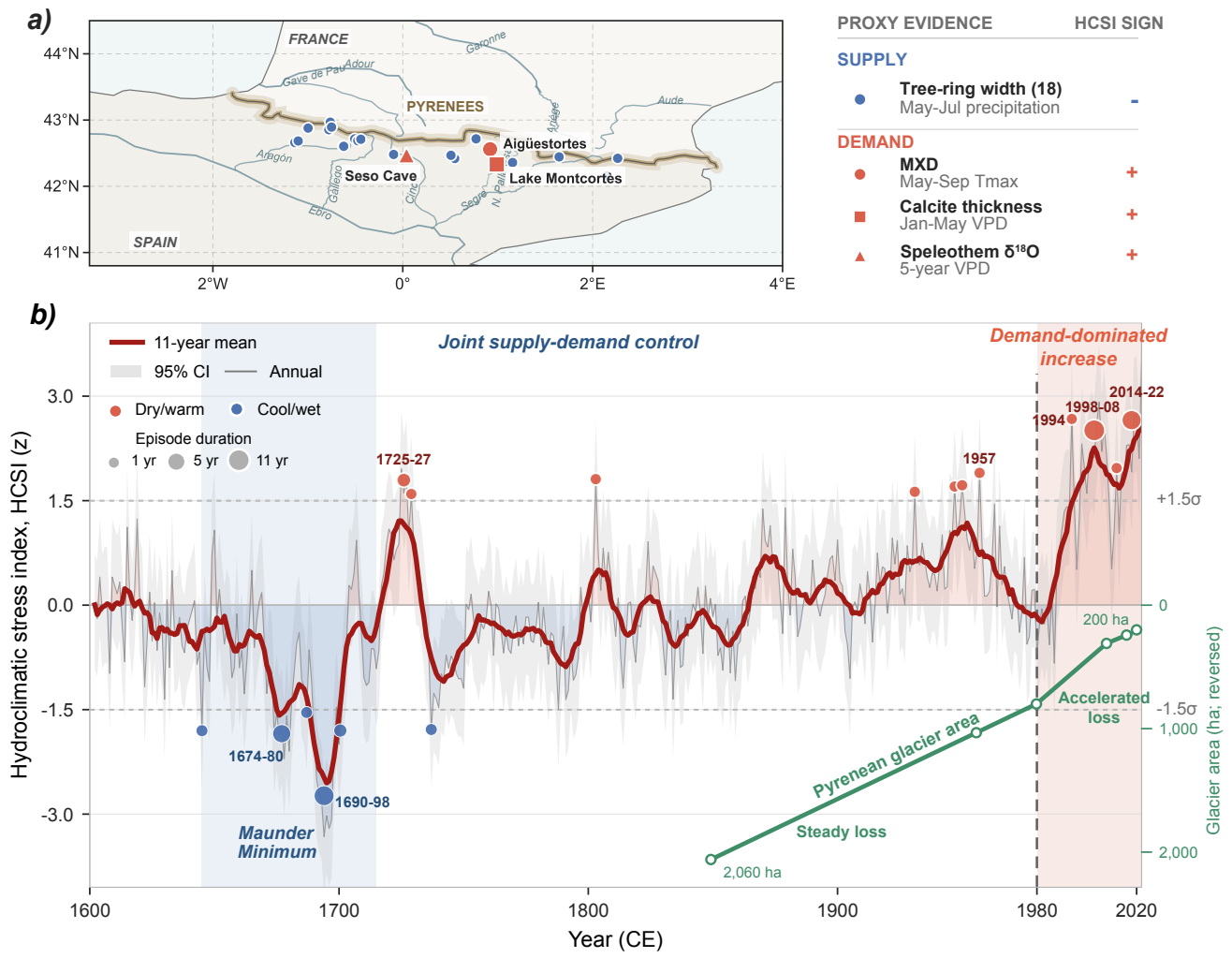

a)
FRANCE
SPAIN
PYRENEES
Aigüestortes
Seso Cave
Lake Montcortès
Gave de Pau
Adour
Garonne
Aragón
Gállego
Cinca
Ésera
N. Pallaresa
Aude
Ebro
44°N
43°N
42°N
41°N
2°W
0°
2°E
4°E
PROXY EVIDENCE
HCSI SIGN
SUPPLY
Tree-ring width (18)
May-Jul precipitation
-
DEMAND
MXD
May-Sep Tmax
+
Calcite thickness
Jan-May VPD
+
Speleothem δ18O
5-year VPD
+
b)
11-year mean
95% CI
Annual
Dry/warm
Cool/wet
Episode duration
1 yr
5 yr
11 yr
Joint supply-demand control
Demand-dominated increase
1725-27
1674-80
1690-98
Maunder Minimum
1957
1994
1998-08
2014-22
Hydroclimatic stress index, HCSI (z)
3.0
1.5
0.0
-1.5
-3.0
+1.5σ
0
-1.5σ
200 ha
Accelerated loss
Pyrenean glacier area
Steady loss
2,060 ha
1,000
2,000
Glacier area (ha; reversed)
1600
1700
1800
1900
1980
2020
Year (CE)